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Part 1: System Idea & Recall

1. UML Diagrams (Descriptions)

1. **Use Case Diagram**
   This diagram shows the interaction between users (actors) and the system. It identifies what actions users can perform in the system. It is useful for understanding system functionality from the user's perspective. It also helps in defining system requirements clearly.
2. **Activity Diagram**
   This diagram represents the step-by-step flow of activities or processes in the system from start to end. It shows decision points and possible paths in the process. It helps in visualizing how the system operates. It is useful for analyzing workflows.
3. **Class Diagram**
   This diagram shows the structure of the system by displaying classes, their attributes, methods, and relationships. It helps in designing the database and system structure. It also shows how objects are related to each other. It is important for system development.
4. **Sequence Diagram**
   This diagram shows how objects interact in a sequence over time, especially how messages are passed between them. It focuses on the order of interactions. It helps in understanding system behavior during operations. It is useful for dynamic processes.

2. System Idea

System Title: Food Menu Ordering System

Short Description:

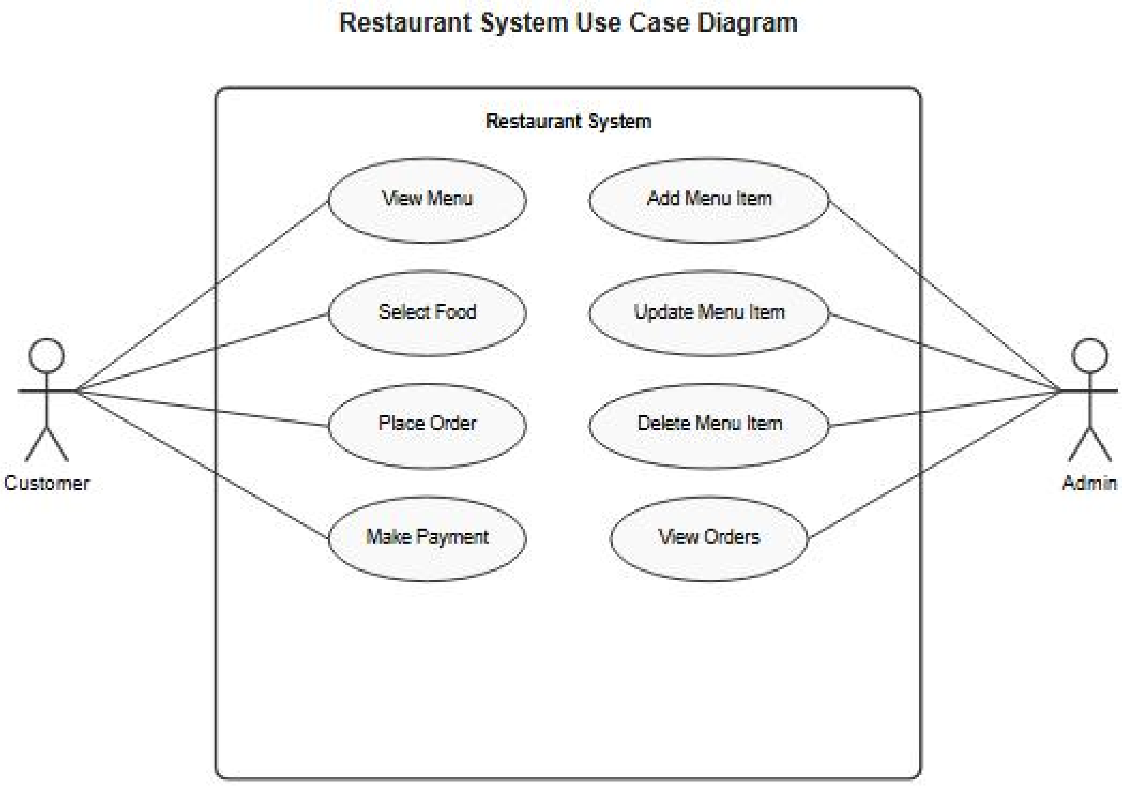
The Food Menu Ordering System is a simple application that allows users to browse food items such as meals, drinks, and desserts. Customers can select items, add them to a cart, and place orders easily. The system also includes a payment feature to complete transactions. In addition, there is an admin panel where menu items can be managed and updated. This system helps make ordering faster, more organized, and convenient for both customers and staff.

Main Features:

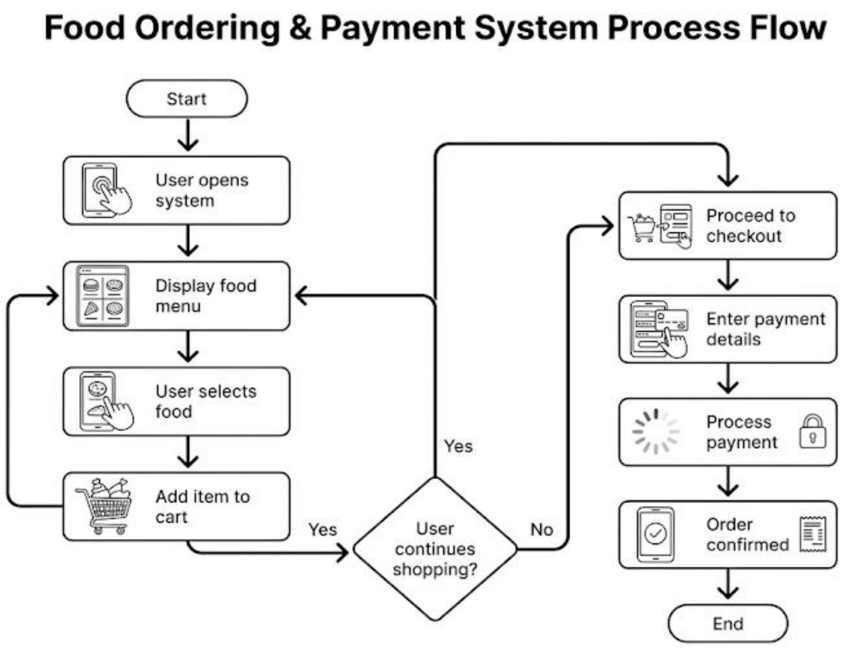
1. View food menu (meals, drinks, desserts)
2. Add items to cart and place order
3. Payment and cashier system
4. Admin panel to manage menu items
5. User login (customer and admin)

Part 2: UML Diagram Creation

1. Use Case Diagram



2. Activity Diagram (Step-by-Step Process)



Part 3: SDLC Application

Planning:
In this phase, the goal of the system is identified, which is to create a Food Menu Ordering System that helps customers order food easily and allows admins to manage menu items.

Design:
The system structure is planned using UML diagrams like use case and activity diagrams to visualize how the system will work.

Implementation:
The system is developed using programming tools (such as Node.js) where features like menu display, ordering, and payment are coded.

Testing:
The system is tested to check for errors, bugs, and to ensure all features like ordering and payment work properly.

Part 4: Reflection

1. What did you learn from creating UML diagrams?
I learned that UML diagrams help in understanding how a system works before building it. It makes the process clearer by showing the flow and interactions.

2. Why is planning (SDLC) important before building a system?
Planning is important because it helps organize the development process, reduces errors, and ensures that the system meets the required goals.